

Dexterous robotic hand with humanoid finger structure and large coverage tactile sensing ability for Human-Robot Interactions

Kuo Liu, Jixiao Liu*, Funing Hou, Dikai Chen, Yidi Zhou, Shijie Guo*

Abstract — Tactile sensory is the most direct source of receiving external physical contact information. Most current robotic hands majorly focus on tactile perception in the fingertip or palm areas for tasks such as object manipulation or recognition. However, it also leads to the missing of tactile information when the robotic hand is engaged with full-hand physical interactions. This paper presents the design of a dexterous robotic hand with large-coverage tactile sensory. Each finger has two active joints driven by DC motors through compact gear transmission a coupling joint driven by a linkage. The entire robotic hand is equipped with flexible iontronic tactile sensors with 492 sensing units covering the fingers, palm, and back of the hand. Human subjects can teach the dexterous robotic hand through tactile information to perform autonomous object grasping, environmental exploration, and physical human-robot physical interaction tasks.

I. INTRODUCTION

It is still far for robotic systems to perform complicated real-world tasks in an elegant manner. The reasons behind this are the lack of human-like robotic sensors and actuators that could naturally feel, recognize, and interact with people or the environment. One of these most typical areas is the robotic hand system.

Currently, robotic hands with underdrive mechanisms demonstrate versatile object manipulation ability by well adapting to object external shapes [1]. When grasping, these robotic hand systems can autonomously adjust contact position and pressure distribution through linkages or pulleys and cables. Therefore, their fingers are flexible to grip a wide range of objects with a single drive source [2]-[5]. Soft robotic hands [6]-[7] have also been a hot academic area in the last decades but are still limited by robust and functional materials and applicable control strategies to solve real-world problems. Still, some humanoid robotic hands own more actuators introduce extra degrees of freedom that ensure better operational functions and flexibility, and synergy among different finger modules [8]-[11]. Although different

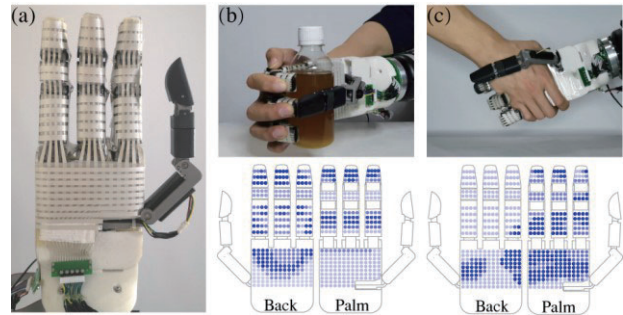


Fig. 1. Dexterous robotic hand with large coverage tactile sensing and its applications (a) The developed robotic hand, (b) Grasping movements by teaching, and the tactile information, (c) Handshake between human and the developed robotic hand, and the corresponding tactile information.

application scenarios place various technical requirements on the robotic hand system, we all must consider the common issue that how to trade-off among factors such as multi-DOFs, structural compactness, hardware costs, etc.

Based on previous research, we argue that fully actuated robotic hands are necessary for more complex manipulation tasks, and there is also a need to simplify the mechanics, reduce the use of sensors, lower the cost of robotic hands and improve their universality.

Tactile sensing is extensively used in dexterous robotic hand systems, along with vision or force sensors [12]-[13]. Visual information can hardly reflect the entire physical status during physical contact between the robotic hand and the environment, especially in the poor ambient light conditions or in the presence of occlusion [14]. Force information from the robotic hand joints would miss the physical contact details such as pressure distribution and stick-to-slip status [15].

The tactile sensing information is crucial feedback for the open-loop robotic hands to analyze the real-time status while making control decisions. It could significantly enhance the manipulation accuracy and control robustness without complicating the mechanical systems of the hand itself. Human hands have no quantitative position or force sensors, but they could perform diverse complex tasks relying on only the tactile sensory. Therefore, large coverage and flexible tactile sensors are more than necessary for the robotic hand to conduct advanced human-robot physical interaction (pHRI) jobs, such as shaking hands with people as non-verbal communications. Partial covered or rigid tactile sensors are still not enough for future natural human-robot interactions [16] - [19].

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This report proposes a dexterous robotic hand with humanoid finger structures and large coverage of tactile sensing. Learned from the human hand structure [20], there are three joints in each finger and two of them are active driving joints (the MCP and PIP joint), as illustrated in Fig. 1 and 2. The third joint (the DIP joint) is coupled with the PIP joint using a linkage. The proposed structure could provide enough DOFs for most manipulation tasks without adding extra mechanical transmissions to the robotic hand. Also, most of the effective working area of the robotic hand is covered by the flexible iontronic tactile sensor, which provides more thorough tactile information during pHRI such as handshaking and environment monitoring such as rapid response when collision. Especially, the tactile sensing on the back of the hand significantly extends the recognition ability of interactive human actions, since most of the pHRI occur through full-hand physical contacts, such as the teaching process and the handshaking demonstrated in the report.

II. DEVELOPMENT OF THE ROBOTIC HAND

A. Mechanical construction

The robotic hand consists of four fingers, the palm, and the wrist section, as in Fig. 1 and Fig 2. The four fingers include the thumb and three finger units. Each finger unit consists of four knuckles and three joints, and they are the metacarpal bone, the proximal knuckle, the middle knuckle, the distal knuckle, the distal interphalangeal (DIP) joint, the proximal interphalangeal (PIP) joint, and the metacarpophalangeal (MCP) joint, in Fig. 2(a). The finger is designed to imitate the human hand movement, with the MCP joint and the PIP joint set as the active joints and the DIP joint set as the coupled drive joint by a linkage, as in Fig. 2(b) and 2(c).

The DIP joint here is a linkage transmission that is more efficient than a gear train when the internal space of the finger structure is very limited. The linkage used here has no potential for grease leakage and requires no pre-tensioning system. In the four-link mechanism, L_1 is the same length as the middle knuckle, and $L_2 = L_4$. The lengths of L_2 and L_4 depends on the position of the joint axis and the size of the space inside the finger unit. Initially, the middle and distal knuckles are in a straight line, and their transmission relationship can be set according to the application scenario. The transmission relationship of the linkage mechanism is not fixed during the movement, so the length of each linkage needs to be optimized. This report sets the transmission relationship between the middle and distal knuckles at 1:1, and $L_1 = 30\text{mm}$, $L_3 = 25.6\text{mm}$, $L_2 = L_4 = 5\text{mm}$.

Two DC motors (DCU10017P06-S002 & PG10C-0064) was used to drive the two active finger joints via worm gears. The drive motor of the MCP joint was assembled inside the palm of the hand together with the metacarpal bone, and the drive motor of the PIP joint was mounted inside the proximal knuckle, in Fig. 2(d). The structure and the driving mechanism of the thumb mechanics are similar to that of the finger unit. There are two active joints at points O and A, in Fig. 2(e). The active joint at point O is driven by a servo, and it controls the thumb move between different angular positions. The active joint at point A is driven by an electric linear actuator, and it controls the thumb flexion and extension.

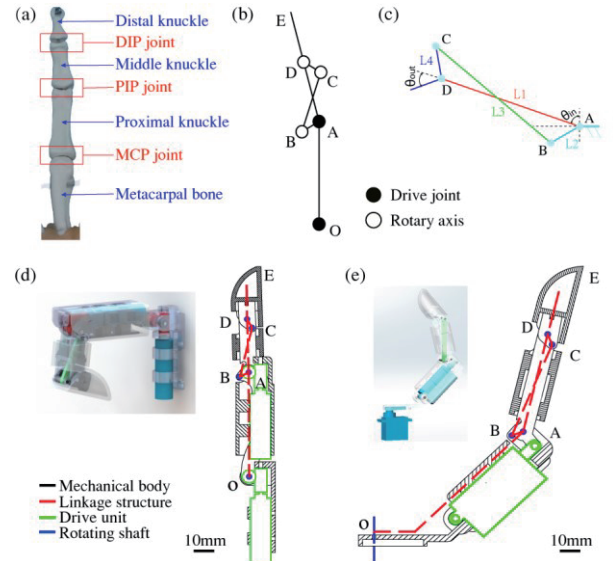


Fig. 2. Mechanical structure of robotic hand (a) The skeleton of a human hand, (b) Sketch of the finger mechanism, (c) Schematic diagram of the coupling drive joint, (d) Mechanics of the finger unit, (e) The mechanics of the thumb.

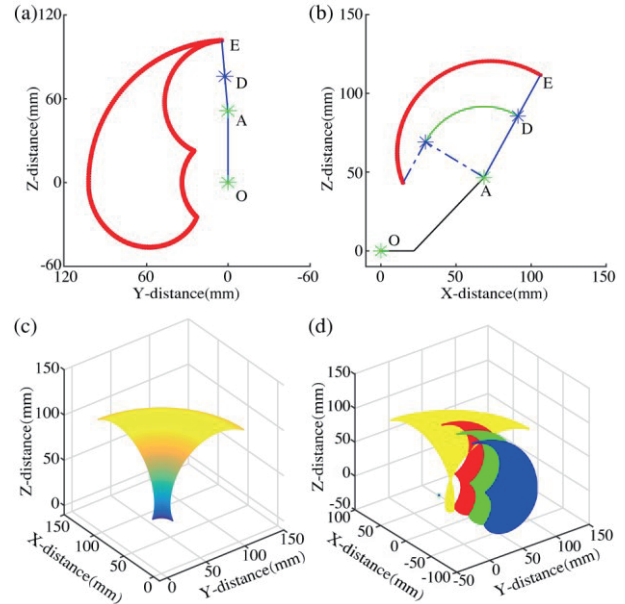


Fig. 3. Work space for robotic hand (a) Working space of finger units, (b) Working space of the thumb (2D), (c) Working space for the thumb (3D), (d) Workspace for the robotic hand.

The length of each knuckle of the robotic hand is based on the dimensions of the human hand: the length of the proximal knuckle is about 50 mm, the length of the middle knuckle is about 30 mm, and the length of the distal knuckle is about 25 mm. Since each finger has two active joints, the fingertip can easily reach wider positions, as depicted in Fig. 3(a). The thumb can flex and extend in the X-Z plane, and it bends up to a maximum angle of 120° , as in Fig. 3(b). The rotation angle around the Z-axis of the thumb is 0° to 90° and the working space of the thumb tip forms a sector in three-dimensional space, as in Fig. 3(c). In the working space of the robotic hand, the working space of the finger unit overlaps with the thumb working space, according to Fig. 3(d). Therefore, the pinching action and grasping action could be achieved through the developed robotic hand.

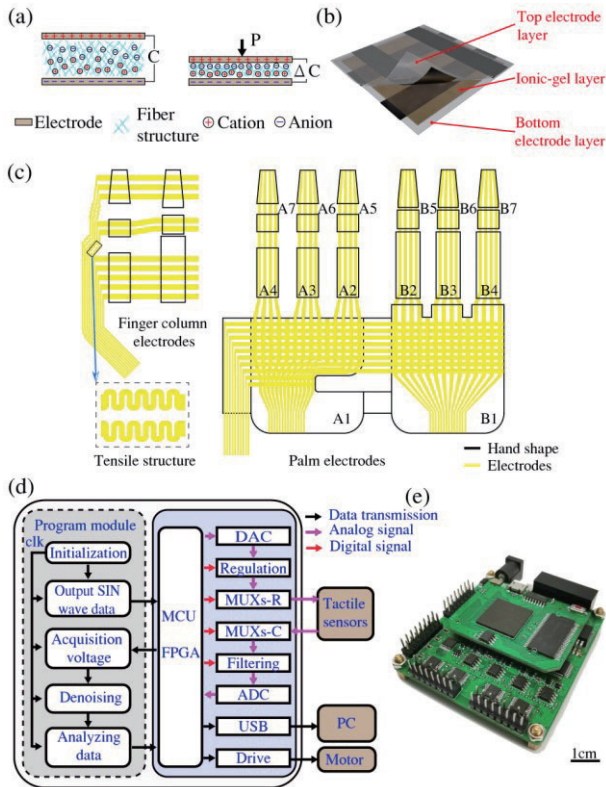


Fig. 4. Tactile sensors and information acquisition circuits (a) Tactile sensor sensing principle, (b) Tactile sensor array structure, (c) Tactile sensor electrodes and sensing areas, (d) Tactile data acquisition and control flowchart of the circuitry, (e) Picture of the actual circuit.

B. Sensors and Information Acquisition

A large coverage iontronic tactile sensor is assembled to the robotic hand system [21]. Unlike conventional capacitive sensors, iontronic tactile sensors consist of arrays of Electric Double Layer Capacitors (EDLCs) at the crossing of the opposing electrodes, as in Fig. 4(a) and 4(b). The sensor is a "sandwich" structure including a top electrode layer, an ionic-gel matrix layer, and a bottom electrode layer. The flexible tactile sensor is shaped and structured based on robotic hand geometries and dimensions, with 492 tactile sensing units in total. The tactile sensor is virtually divided into 14 areas, as illustrated in Fig. 4(c). The tactile sensing areas in the palm are denoted by "Ax", and the areas in the back of the hand are denoted by "Bx". Since the MCP joint, the DIP joint, and the PIP joint would stretch the covering sensors when the finger is bent, the connecting parts of the sensor longitudinal electrode at the joint positions are made with serpentine structures to increase the stretchability preventing sensor electrode failure during the hand finger movement.

Except for the side area of the palm and the fingers, the tactile sensor covers most of the effective area of the developed robotic hand. Since the thumb works in a supporting role in most of the manipulation and interactive tasks and it also owns large rotation angles that the tactile sensor can hardly fit, therefore, in the demonstrated prototype there is no tactile sensor assembled to the hand thumb.

The dexterous robotic hand needs to respond quickly during operation, it requires the system to collect and analyze the tactile information, and feedback in real-time. A system

integrated data acquisition and robot control is built with FPGA as the MCU, as in Fig. 4(d) and 4(e). The vertical and horizontal electrodes of the tactile sensor are connected to the row and column interfaces of the circuitry. To improve the acquisition speed, the four-column interfaces are opened simultaneously to collect, modulate, and filter the collected data. Finally, the pressure applied to each tactile unit is proportional to the voltage value [21], and the collected voltage values are converted into 12bits data by ADC and normalized, as in Fig. 4(d). The data of each tactile unit can be stored individually to make a pressure image or summed for all tactile units in different areas as a threshold range for the grasping action. For example, B2 has 20 tactile units, so the maximum threshold range for that area is 20. The circuitry is palm-size and can be installed inside the wrist part of the robot. The circuit also features motor protection and USB data transfer functions.

C. Control methods

Based on the tactile information and motor current information, the motion control of the robotic hands can be divided into three modes: manual control mode, tactile-based teaching mode, and autonomous gripping mode, as in Fig. 5.

Manual control mode: In this mode, tactile information is only plotted for display and data record, and not involved in

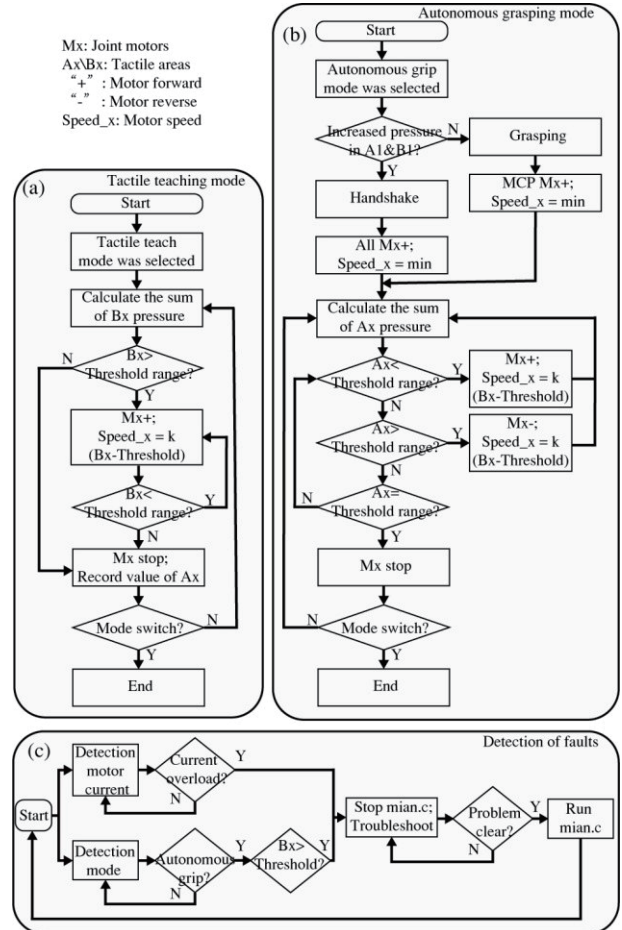


Fig. 5. Control flow chart (a) Flow diagram of the robotic hand control program, with "Mx" indicating each knuckle motor, "+" and "-" signs indicating forward and reverse motor rotation, respectively, and Ax and Bx corresponding to the zones delineated in Fig. 4(c), (b) Interruption detection procedure in the event of a fault.

the control of the robotic hands. The data are mainly used for robotic hands debugging and performance testing or controlling the robotic hands to show certain gestures and other manual adjustments.

Autonomous grasping mode: As described in Fig. 4(c), the tactile sensor is virtually divided into several sub-areas (such as Ax and Bx) and each sub-area contains certain sensing units. The tactile data from the sensors in each sub-area is mathematically summed which is used as the feedback value to judge the real-time contact status in that sub-area. Before the autonomous grasping starts, the robotic hand system is notified with the information of the target object, then the robotic hand selects the corresponding data from the database. The robotic hand can grasp objects autonomously according to the tactile information recorded by the teaching mode. The tactile information (the data set of

distributed contact pressure) from the teaching mode is used as the target threshold range when the robotic hand grasps objects or handshakes with human where the contact pressure varies:

- (1) If the pressure variation is not obvious or appears from all the proximal knuckles, the grasp action will be performed;
- (2) If the pressure variation appears on the A1 and the B1 areas, the handshake action will be performed.

When the real-time contact pressure in all areas reaches and hold within the threshold range, all motor movements stop.

III. EXPERIMENT

The developed robotic hand was mounted to a six-axis robot arm (Zhiyin Robotics (Beijing) Inc, China) for the

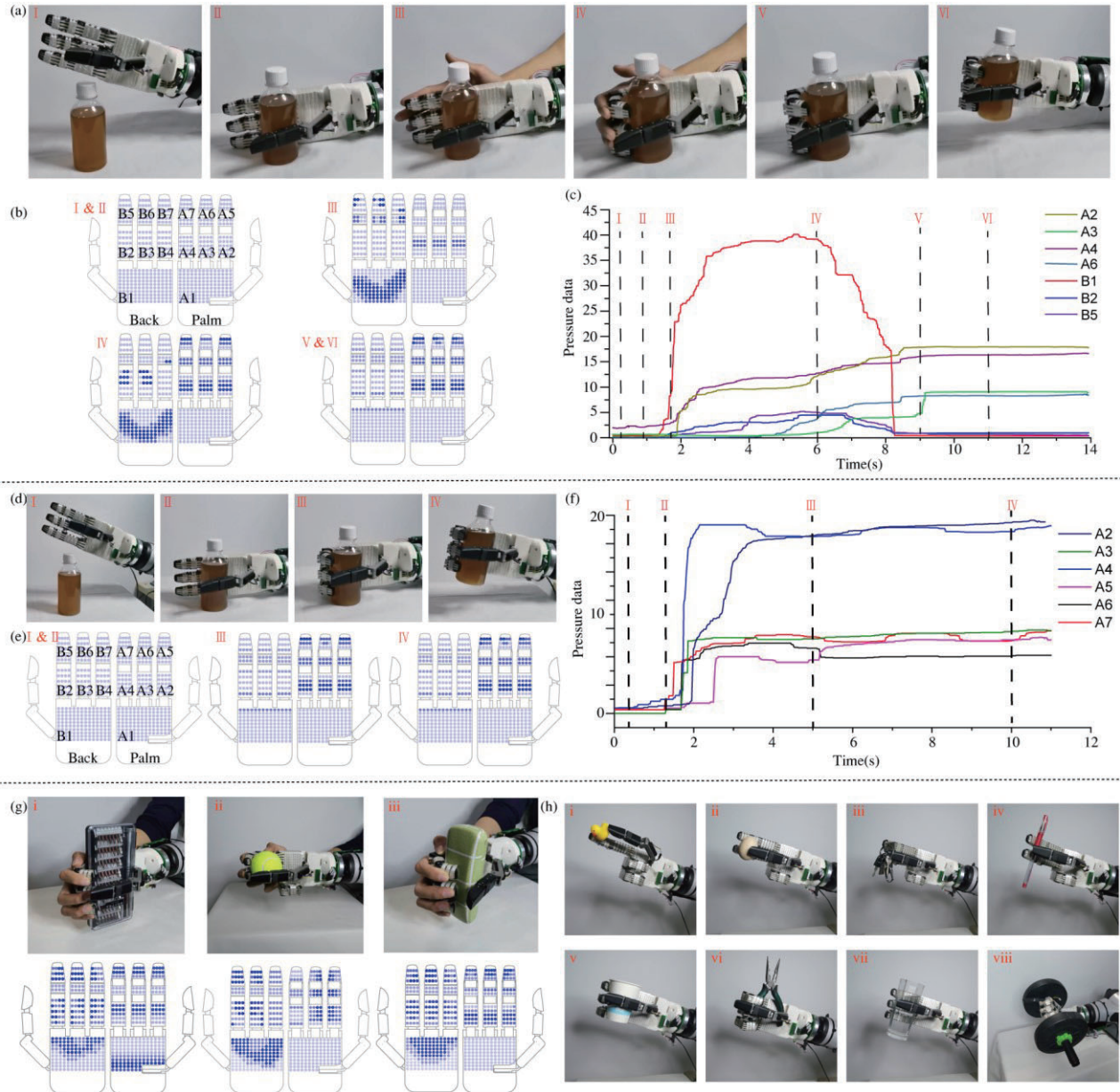


Fig. 6. Tactile teaching and autonomous grasp experimental (a) Experimental procedure and data of tactile teaching, (b) Experimental procedure and data of autonomous grasp, (c) Tactile teaching experiments with different objects and pressure diagrams at corresponding moments, (d) Grasping experiment of different objects.

experimental test. Both object grasping and physical human-robot interactions were separately tested using the developed prototype and methods as depicted above.

A. Tactile teaching experiment

Firstly, a teaching grasp experiment is conducted based on tactile information, and the pressure data variation is demonstrated in Fig. 6(a)-(c). As in Fig. 6(a)-(c), at the moment I and II, before the experiment, a human operator controlled the arm to drive the robotic hand to reach the top of the object and move down along the contour of the object until the appropriate position for grasping. Then, the thumb rotated to 90°. No pressure was applied in this progress. As in Fig. 6(a) to 6(c), at moment III, the tactile teaching process started, the human operator contacted the opisthenar, and the pressure on B1 triggered the teaching grasp mode. Then, the triggered area Bx indicated the corresponding Mx motor to bend the fingers, which made the robotic hand grasp the target object. As in Fig. 6(a)-(c), at the moment IV, signals appear in the palm areas and the human operator needs to determine whether the robotic hand has a stable grasp and whether to stop applying force. As in Fig. 6(a)-(c), at the moment V, the grasp state was spied by the human operator until the object was no obvious slide and deformation. After that, the subject detached the robotic hand, and the arm moved the grasped object to test whether the grasp is stable. As in Fig. 6(a)-(c), at the moment VI, although the object was stable when the arm moved, the pressure data vibratory, resulting from the object gravity distribution changing, could be detected. If the object did not slide, the tactile data would be recorded and added to the database for the action autonomous grasping experiment.

Due to the large number of areas, only the regions with obvious change characteristics are shown in the figure. The trend of variation in the distal knuckle was similar to that of A6, the pressure variation in the proximal knuckle of the opisthenar was similar to that of B2, and the pressure variation in the distal knuckle of the opisthenar was similar to that of B5.

This report also performed the tactile teaching test for some other daily objects, and showed the pressure images of some of them at steady state as Fig. 6(g). For larger and

regularly shaped objects, as in Fig. 6(g) of i, ii, iii, and as in Fig. 6(h) of v and vii, the human operator can place them on the experimental table for the teaching grasp experiment; for smaller and irregularly shaped objects, as in Fig. 6(h) of i, ii, iii, iv, vi and viii, the human operator needs the cooperation of another human operator to pose the objects in a suitable position. The function of the tactile teaching experiment is to get the threshold range of force applied by the robotic hand to each experimental object through multiple trials and to create a database waiting for the call of the autonomous grasping or interaction experiment

B. Object grasping and manipulation

During the autonomous grasp experiment, the robotic hand performed grasp action without the operator. As in Fig. 6(d) to 6(f), at the moment I and II, before the experiment, the human operator controlled the arm to drive the robotic hand to reach the top of the object and move down along the contour of the object until the appropriate position for grasping. Then, the thumb was rotated to 90°. No pressure was applied in this progress. After the action, the human operator notifies the robotic hand of the target object, then the robotic hand selects the corresponding data from the database. As in Fig. 6(d) to 6(f), at the moment III, in the beginning, the robotic hand performed grasp action without the human operator. The MCP moved until the proximal knuckles contacted the object. Then, the PIP and the DIP were driven simultaneously by the linkage mechanism. During this progress, the speed was proportional to the deviation between the current and the target pressure data. When the sum of each area's pressure data was close to the recorded data, all the joints stopped, this indicates that the robotic hand has a stable grasp. As in Fig. 6(d)-(f), at the moment IV, the arm drives the robotic hand to lift objects, because the center of gravity of the object is changed, the force on the robotic hand will be changed accordingly. If the pressure data of a certain area Ax leaves the threshold range, the corresponding motor Mx would be adjusted until the pressure data of all areas were kept within the threshold range. During this progress, the opisthenar and A1 has no obvious variation, so only the pressure variation in A2 to A5 was shown.

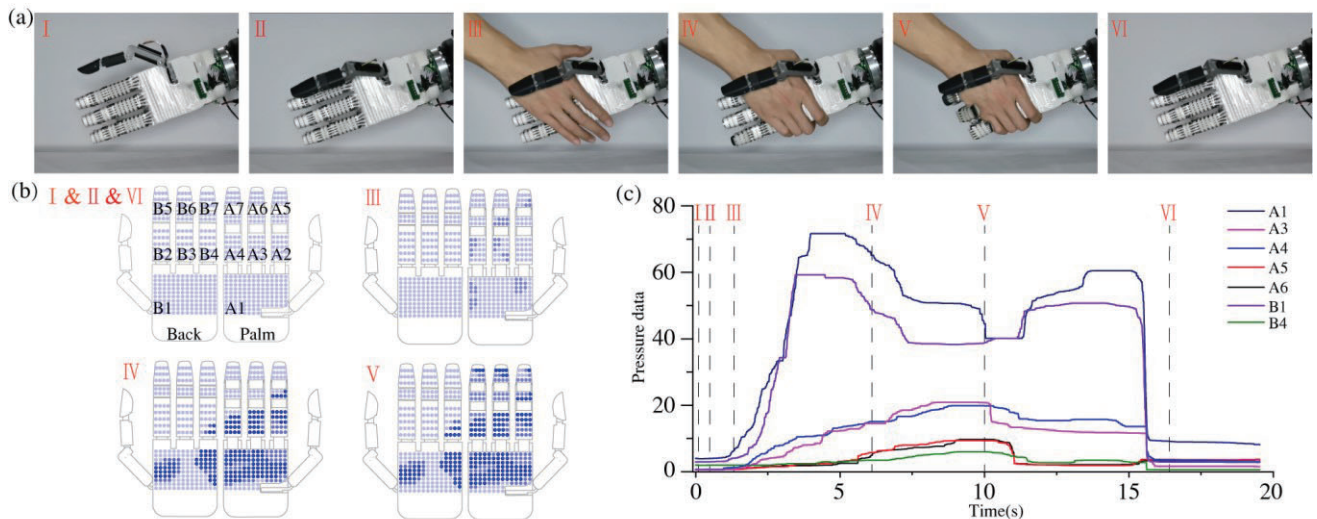


Fig. 7. Interaction experiments (a) Time-series diagram of the handshake process; (b) Handshake process pressure profile; (c) Pressure data during operation.

C. Physical Human Robot Interaction experiments

This report considers the security of human-robot interaction in the same way as ensuring that objects are not deformed when grasped; both simply require controlling the pressure at all contact points. Thus, before the interaction experiment, the human operator simulation the handshake process in tactile teaching mode and record the pressure threshold range of all areas.

As in Fig. 7, at the moment I and II, before the experiment, the human operator controls the arm, driving the robotic hand becomes a downward sloping handshake position. Then, the thumb was rotated to 30°. No pressure was applied in this progress. After the action, the human operator notifies the robotic hand of the target object, then the robotic hand selects the corresponding data from the database. As in Fig. 7, at the moments III and IV, when the subject held the robotic hand, the pressure data of the B1 and the A1 areas triggered the autonomous handshake mode. Unlike the progress of grasping an object, when the robot grasping the human hand, all the motors moved simultaneously and adjusted their speed individually. As in Fig. 7, at the moment V, when the pressure data of all the areas in all areas remain within the threshold range and maintained for a period of time, the handshake was finished. Then, the robotic hand posture was reset, and the subject detached the hand.

Handshake is a basic movement in non-verbal communications between people, and it has been a challenge in physical human-robot interactions. Comparing the tactile images of IV and V, it is known that the pressure was felt in B1, A1 and some knuckles immediately after the subject grasp the robotic hand. Until the robotic hand completes the handshake, the pressure changes in these areas are small and mostly due to the shaking or adjustment of the hand during the handshake. The main change in this process is in the middle and distal knuckle part of the robotic hand. During the handshake, the distal and middle finger joints need to cover the human hand and keep the pressure values in all areas within the threshold range. During the handshake, the rotation of the MCP joint is minimal, while the movement of the PIP joint requires more attention.

IV. CONCLUSION

This report presents a dexterous robotic hand with humanoid finger structure and large coverage tactile sensing ability. However, it is only a prototype test now. This report attempts to use tactile sensors to detect information and perform tactile teaching experiment, autonomous grasp experiment and human-robot interaction experiments. In future work, we will increase the functionality of the thumb, improve the resolution and coverage of the tactile sensors, build a complete database, and add artificial intelligence algorithms to train the control strategy of the robotic hand.

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